

Samwel Waweru

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🌐 [samkamau81](https://github.com/samkamau81)

🌐 [samkamau81.github.io](https://github.com/samkamau81.github.io)

EDUCATION

Columbia University

New York, USA

M.S. Artificial Intelligence (Robotics and Perception Concentration)

- Research focus on on-device intelligence, efficient inference, multimodal AI systems, computer vision, robot learning, retrieval systems, and embedded machine learning under resource-constrained environments.
- Relevant Coursework: Machine Learning, Computer Vision I & II, Embedded AI, Natural Language Processing, Robot Learning, Optimization for Intelligent Systems.

Jomo Kenyatta University of Agriculture and Technology

Nairobi, Kenya

B.S. Mechatronic Engineering (Honors)

- Ranked **1st out of 18** capstone projects for developing an autonomous AI-enabled Delta Laser Weeding Robot integrating real-time computer vision, embedded robotics, and low-latency control systems.
- Relevant Coursework: Embedded Systems, Robotics, Control Systems, Data Structures, Computer Architecture, Signal Processing, Applied Optimization.

WORK EXPERIENCE

AI Solutions Expert, EdTech Hub Kenya

Kenya / UK Hybrid

Oct 2025 – Present

- Led development of the Golden Record System to resolve data inconsistencies across national education system using Python, SQL, Scikit-learn and Databricks.
- Architected a hybrid retrieval-augmented generation (RAG) system integrating SQL analytics with FAISS and retrieval pipelines to generate grounded, source-traceable policy responses using Python, PyTorch, REST APIs, and modern LLM infrastructure.
- Provided technical counsel on AI adoption, trustworthy AI systems, scalable infrastructure, and digital transformation initiatives to senior government officials and the Office of the Cabinet Secretary for Education.

Machine Learning Engineer, Kaggle BIPOC Program

USA, Remote

Dec 2022 – Oct 2025

- Developed and deployed machine learning systems emphasizing scalable inference workflows, reproducible experimentation, and reliable API-based deployment architectures using Python, PyTorch, TensorFlow, and FastAPI.
- Engineered containerized ML pipelines with Docker and CI/CD automation workflows to support low-friction experimentation, deployment reproducibility, and maintainable model iteration.
- Mentored four fellows on NLP systems, experimentation workflows, optimization strategies, and multimodal AI model evaluation.

AI Research Associate, The University of Manchester

Manchester, United Kingdom

2021

- Developed natural language processing models using Hugging Face Transformers and BERT architectures to perform sentiment analysis on Twitter data collected through the Twitter API using Python. [GitHub Link](#)
- Winner of the Machine Learning Hackathon for a project investigating the effects of COVID-19 on mental health, applying data-driven analysis and machine learning techniques to identify trends and insights from public data sources.

SELECTED PROJECTS

Golden Record, 2026 (AI-Enhanced Educational Data Integrity Platform)

- Designed and implemented AI-assisted entity-resolution and validation pipelines for large-scale educational datasets using Python, SQL, Scikit-learn, and Databricks to improve identity consistency across fragmented national records.

- Developed a hybrid RAG pipeline combining SQL-based analytics with FAISS and BM25 semantic retrieval over policy documents to generate grounded and traceable insights for education policymakers using Python, PyTorch, and LLM orchestration workflows.

Delta Laser Weeder, 2025 (Real-Time Edge AI Robotics)

[GitHub Link](#)

- Built an autonomous embedded robotics platform integrating computer vision, precision laser targeting, inverse kinematics, and real-time motion control for intelligent weed removal.
- Optimized YOLOv11 inference pipelines for Raspberry Pi deployment using lightweight edge-computing strategies, embedded Linux environments, and low-latency perception workflows.
- Integrated embedded perception outputs with Arduino-based actuation pipelines to support deterministic robotic control under real-world field constraints.

EmbedMind Studio, 2025 (AI Agent Development Platform)

[GitHub Link](#)

- Developed an embedded AI agent platform enabling deployment of intelligent automation workflows to edge devices and IoT systems.
- Designed modular orchestration pipelines integrating sensors, actuators, lightweight LLM workflows, deployment automation, and cloud-connected monitoring systems.

FinGPT, 2024 (Financial Language Model Platform)

[GitHub Link](#)

- Developed backend infrastructure for a domain-specific financial language model platform using FastAPI with strongly typed Pydantic schemas and scalable API design principles.
- Engineered asynchronous inference services integrating financial text-processing workflows, structured validation layers, and maintainable request-response architectures.

Kilimo-AI, 2023 (LLM-Assisted Smart Irrigation System)

[GitHub Link](#)

- Developed an intelligent irrigation platform integrating lightweight LLM-assisted decision making with IoT-based agricultural automation systems.
- Engineered Python-C++ interoperability modules translating natural-language user interactions into embedded device-level irrigation commands for low-power agricultural controllers.

COMMUNITY INVOLVEMENT

County Coordinator / Master Trainer, Raspberry Pi Foundation

Kenya

Apr 2024 – Aug 2025

- Established 10 Code Clubs and trained over 250 learners in programming, robotics, embedded systems, and applied AI engineering.

PUBLICATIONS

- **Samwel Waweru** (Lead Author), Ray Njoroge. *Edge-Optimized YOLOv11-Based Real-Time Weed Detection for Autonomous Precision Agriculture*. Accepted at **IEEE/RSJ IROS 2025, China** (Oral Poster).
- Jackson Mutembei, **Samwel Waweru** (Contributor). *A Biodiversity Conservation Protocol for Fire Prevention using Drone Technology*. **University of Birmingham, 2022, UK** (Oral Poster).

ADDITIONAL INFORMATION

Research Interests: On-device intelligence, efficient inference, multimodal AI systems, embedded machine learning, retrieval systems, robotics, optimization, computer vision, and human-centered AI systems.

Languages: English (Fluent), Kiswahili (Native)